SegCR: A Multimodal and Multitask Complementary Fusion Network for Remote Sensing Semantic Segmentation and Cloud Removal

Shicheng Wu, Jinbiao Zhu, Yanlin Gu, Wenqi Han, Wen Jiang, *Member, IEEE*, and Jie Geng, *Member, IEEE*

***Abstract*—The synthetic aperture radar (SAR) can provide complementary information to optical images due to its insensitivity to atmospheric conditions, making optical–SAR fusion semantic segmentation a popular research topic. However, existing optical–SAR fusion methods overlook cloud interference in real-world scenarios, leading to the suboptimal performance in practical applications. To address the performance degradation of existing optical–SAR fusion semantic segmentation methods under cloud interference, we propose SegCR, a multimodal and multitask framework that leverages the complementary information of SAR images to reduce the negative influence of clouds in optical images. Specifically, we extract a cloud impact map using the frequency differences between optical and SAR features, which represents the extent of cloud interference for every pixel in the optical image. Next, we introduce an SAR-to-OPT translation subtask, leveraging SAR features to produce optical simulated features that supplement the missing information in the cloud-affected optical features. Then, based on the cloud impact map, we perform a targeted complementary fusion of optical and SAR features. Finally, we design a multitask learning framework that simultaneously performs the semantic segmentation and cloud removal, enabling the high-level semantic understanding task and the low-level vision task to enhance each other through multitask learning. Extensive comparison experiments on two publicly multitask remote sensing datasets reveal that our proposed SegCR outperforms the existing state-of-the-art (SOTA) methods. In addition, ablation experiments have confirmed the effectiveness of the proposed module and the multitask learning framework.**

***Index Terms*—Cloud removal, multimodal fusion, multitask learning, remote sensing, semantic segmentation.**

I. Introduction

SEMANTIC segmentation of optical remote sensing images is of vital significance in the field of Earth science research [1]. Unlike image classification tasks, it assigns fine-grained labels to every single pixel within an image. As a result, it offers a more thorough and comprehensive depiction of the characteristics of the land surface. This functionality is of utmost importance in a diverse range of applications, such as natural resource surveys [2], urban management [3], and disaster emergency response [4]. As the demand for higher accuracy in land cover mapping increases, traditional semantic segmentation methods, such as support vector machines (SVMs) [5] and conditional random fields (CRFs) [6], are no longer sufficient. With the rapid progress of deep learning technologies, an ever-growing quantity of deep learning approaches have been utilized for optical image semantic segmentation tasks and accomplished commendable results [7]. When collecting surface information, optical remote sensing satellites will unavoidably be affected by atmospheric clouds, resulting in spectral data loss in specific regions. This disruption hinders the precise classification of land cover types and fails to meet the demands for quality and quantity of remote sensing research and applications. For instance, in regions like Southeast Asia and the Atlantic coast, cloud cover persists almost year round [8], making it challenging to achieve timely and accurate land cover mapping.

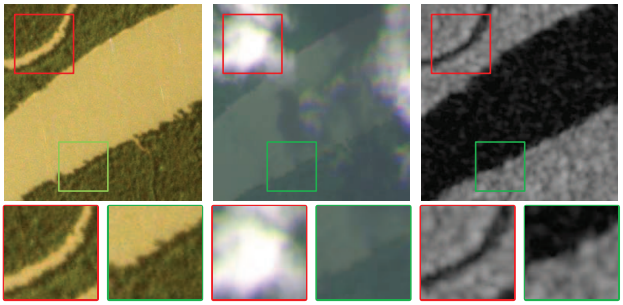


Fig. 1. Illustrative example showing how cloud occlusion disrupts the consistency between optical and SAR images. (a) Cloud-free optical image. (b) Cloudy optical image. (c) SAR image. The regions within the green boxes highlight areas where optical and SAR images exhibit consistent features, while the regions within the red boxes demonstrate conflicts between the optical and SAR images due to cloud occlusion, which may cause confusion in deep learning models.

In recent years, due to the rapid advancement of remote sensing imaging technologies, it has become available to obtain high-resolution remote sensing data from various sources, including multispectral [9], hyperspectral [10], and synthetic aperture radar (SAR) [11]. SAR, based on microwave radar waves, can actively emit signals and receive reflected waves, making it independent of external light sources and unaffected by cloud cover or lighting conditions [12]. Owing to

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its imaging mechanism, SAR serves as a valuable supplement to optical images since it can offer exclusive information absent from optical data [13]. As a result, the fusion of optical and SAR images for semantic segmentation has become a focal point in the remote sensing image analysis [14]. However, current optical–SAR fusion approaches overlook the presence of clouds in optical images [15]. As shown in Fig. 1, the correlation and consistency between optical and SAR images are interfered by clouds, leading to conflicts that may misguide the learning process. Therefore, networks that are trained on cloud-free optical images may not achieve the desired performance when performing multimodal fusion semantic segmentation with cloudy optical images and SAR images [16].

To enhance the semantic segmentation under cloud cover conditions, a two-stage approach utilizes SAR images to restore optical images obscured by clouds, followed by multimodal remote sensing image scene understanding [17]. Ideally, the cloud-removed optical images should exhibit characteristics similar to cloud-free optical images. However, this paradigm introduces a multimodal cloud removal network, significantly increasing network complexity and computational load. In addition, cloud removal methods often prioritize visually appealing results for human observation, which may inadvertently compromise semantic information and discriminative features beneficial for semantic segmentation. Consequently, the upstream cloud removal task may not necessarily improve the downstream semantic segmentation performance. An alternative approach involves direct multimodal fusion of cloudy optical images and SAR images during the training process, allowing the neural network to learn specific patterns of clouds [18]. By omitting the cloud removal step, this method preserves the important semantic information within the optical images. However, the presence of clouds can lead to incomplete or distorted semantic information within the optical images, disrupting the correlation and consistency between optical and SAR images. This introduces unavoidable confounding information into the neural network's learning process, negatively impacting the optical–SAR multimodal fusion.

In order to solve the foregoing issue, we propose a novel semantic segmentation and cloud removal network named SegCR. Specifically, we first use a dual-branch backbone network to extract features from the cloudy optical image and the SAR image. Then, we introduce a frequency difference cloud mapping (FDCM) module, which employs Haar discrete wavelet transform (DWT) [19] to extract the low- and high-frequency components of optical and SAR features. By comparing the differences in the frequency components, we generate a cloud impact map, which represents the degree of cloud interference on each pixel in the optical features. Next, to compensate for the missing information caused by cloud occlusion, we propose an SAR-to-OPT feature transfer (SOFT) module. This module utilizes the information from SAR features to generate corresponding optical information for the affected regions. Specifically, it uses the multilevel latent optical features extracted through the auxiliary optical image reconstruction subnetwork [20] to guide the SAR feature transfer subnetwork in generating the realistic optical simulated features. Notably, the SOFT module is trained before the other parts of the network, and once the training is completed, its parameters are frozen. Afterward, we introduce a cloud-guided complement fusion (CGCF) module, which uses the cloud impact map generated by FDCM to selectively complement the areas of the original optical features that are affected by cloud occlusion, resulting in optical corrected features. These optical corrected features are then fused with the SAR features to obtain fused semantic features. Finally, we design a multitask learning architecture, where the obtained multilevel optical corrected features and fused semantic features are fed into the semantic segmentation decoder and the cloud removal decoder, respectively. This brings about the final outputs: land cover maps and cloud-removed optical images.

The contributions of our research are outlined as follows.

1) We present a multimodal and multitask network framework capable of simultaneously performing semantic segmentation and cloud removal tasks. By fostering collaboration between high-level semantic understanding tasks and low-level visual tasks, our framework achieves state-of-the-art (SOTA) performance for both the semantic segmentation task and the cloud removal task.
2) We propose an FDCM module, which generates a cloud impact map by comparing the low- and high-frequency component difference between optical and SAR features. The cloud impact map represents the degree of cloud interference on each pixel in the optical features, effectively guiding the subsequent CGCF module to selectively preserve cloud-free optical features while correcting cloud-affected optical features.
3) We introduce an SOFT module, which utilizes the multilevel feature guidance through an auxiliary optical reconstruction task. This approach enables the SAR feature transfer subnetwork to generate realistic optical spectral information using SAR scattering information, effectively compensating for the spectral information loss due to cloud cover.

II. Related Works

A. Multimodal Remote Sensing Semantic Segmentation

The semantic segmentation has been an enduring research issue in the domain of remote sensing [21]. Before the advancement of deep learning methods, traditional machine learning approaches were commonly used [22]. Kasetkasem et al. [23] proposed a joint classification method based on the Markov random field (MRF) model. This method aims to align two images at the same time and acquire a land cover map for the observed scene [23]. Since these methods depend on manually designed features like color, texture, and shape, they are unable to capture more valuable high-level semantic information, resulting in unsatisfactory semantic segmentation accuracy.

As deep learning techniques have been developing at a rapid pace, an ever-growing number of neural networks have been utilized for multimodal remote sensing images semantic segmentation [24]. Ma et al. [25] addressed the

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limitations of existing methods in local–global context modeling and representative feature learning by proposing a multilevel multimodal fusion framework named FTransUNet. By combining CNN and ViT in a unified architecture, this approach effectively merges shallow and deep multiscale features. Xie et al. [26] proposed an Ebbinghaus curve-guided multimodal land cover classification network (ECNet), which preserves essential features while eliminating redundancies in the high-dimensional manifold structure, thereby mitigating overfitting caused by redundant multimodal features. Fuller et al. [27] addressed the challenge of learning effective representations from large scale, sparsely labeled, and spatially aligned multimodal images by proposing CROMA—a self-supervised framework that jointly optimizes contrastive and masked autoencoding objectives. This method employs separate encoders for optical and SAR inputs, as well as a multimodal encoder, to learn both unimodal and fused features via cross-modal contrastive learning and multimodal reconstruction.

The aforementioned methods have made improvements in multimodal remote sensing semantic segmentation from various aspects, achieving good results on laboratory datasets. However, they do not consider the presence of clouds in real-world scenarios, meaning that directly applying these methods in practical situations may not yield satisfactory results.

B. Optical–SAR Cloud Removal

In remote sensing research, optical images are easily affected by atmospheric clouds. As some regions of the image are affected by clouds, cloud noise can severely impact the quality of the data. Therefore, it becomes particularly important to remove the cloud occlusion while minimizing the loss of original valid information [28].

Since SAR images can penetrate atmospheric phenomena such as clouds, rain, and snow using microwave signals to capture the reflected information from ground objects, SAR images have become an ideal choice for addressing the cloud occlusion problem [29]. Xu et al. [30] proposed a global–local fusion-based cloud removal algorithm (GLF-CR), which leverages the embedded complementary information in SAR images. The algorithm ensures that, during global fusion, the structural consistency is maintained between the restored areas and the other cloudless regions. At the same time, it creates dependable texture details for the areas that were originally missing in local fusion. Ma et al. [31] addressed the problem of image blur and texture loss in cloud removal tasks based on SAR–optical image fusion. They proposed SCT-CR, a synergistic convolution-transformer network that integrates a cloud-shadow attention mechanism, a novel SynConv module for adaptive fusion, and a transformer encoder to enhance global feature learning. Zhang et al. [32] addressed the challenge of removing clouds from optical remote sensing images by leveraging the complementary properties of SAR data, which are insensitive to cloud cover. They proposed DMDiff, a dual-branch multimodal conditional diffusion model that integrates SAR and optical features through a cross-attention fusion module and introduces an image adaptive prediction strategy to better handle spatial–spectral complexity.

Although these methods achieve good visual results in cloud removal, they inevitably suffer from the loss of the semantic information and discriminative features in the optical images that are beneficial for semantic segmentation. As a result, using these cloud-removed optical images for downstream image understanding tasks may not yield ideal outcomes [16].

C. Multitask Learning

Multitask learning has been widely applied in the image understanding task. It is widely recognized that high-level semantic tasks and low-level visual tasks often exhibit semantic or structural complementarity, thus, joint modeling of such tasks can facilitate the learning of more generalizable and robust intermediate representations. However, due to different optimization objectives, the auxiliary and primary tasks may not achieve their respective optimal performance. Therefore, it is necessary to introduce a certain degree of task independence to mitigate potential negative interference between them [33].

Cai et al. [34] tackled the challenge of poor segmentation performance on low-resolution target images caused by significant spatial resolution shifts in the unsupervised domain adaptation. They proposed a multitask domain adaptation network (DARSNet) for remote sensing image semantic segmentation with the aid of super-resolution. In this framework, the domain adaptation branches for semantic segmentation and super-resolution share the same encoder to extract domain-invariant features, while employing distinct decoders and discriminators to perform cross-domain semantic segmentation and super-resolution tasks, respectively. Zhao et al. [35] tackled the challenge that real-world point cloud data often contain extraneous noise and irrelevant points, which pose difficulties for downstream tasks. They proposed a framework that utilizes a shared PointNet encoder and three branch networks to perform point cloud denoising, single-object segmentation, and completion. The three branches share features, enabling the model to learn cross-task representative features. For each branch, specific loss functions were designed.

As evidenced by the aforementioned methods, isolating the auxiliary and primary tasks through the use of separate encoders and task-specific loss functions can effectively mitigate the implicit interference of the auxiliary task on the main task during the learning process. This strategy enables the model to achieve satisfactory performance on both tasks simultaneously.

III. Proposed Method

A. Overview

Fig. 2 demonstrates the overall framework of the proposed SegCR. First, we employ two parallel Segformer-B2 [36] backbone networks to extract deep features from cloudy optical images and SAR images, respectively. Then, the optical and SAR features are fed into the FDCM module, where the low-frequency and high-frequency differences between the two are compared with extract the cloud impact map, which denotes the extent of interference by clouds for every pixel in the optical image. Next, in the SOFT module, the optical image reconstruction task is utilized to guide the

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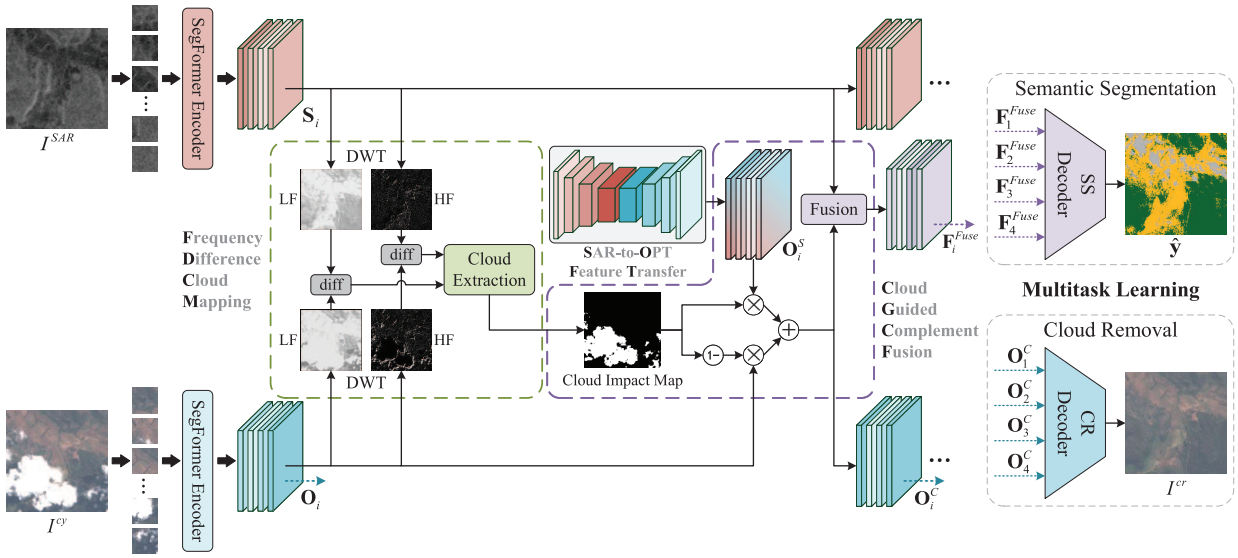


Fig. 2. Overall architecture of SegCR. The input consists of SAR images and cloudy optical images. The green dashed box represents the FDCM module, while the purple dashed box represents the CGCF module. The SOFT module is positioned between them. After obtaining the optical corrected features and fused semantic features, these are fed into the cloud removal decoder branch and the semantic segmentation branch for multitask collaborative learning. Finally, the cloud-removed optical images and the land cover maps are obtained.

transfer of SAR scattering information to optical spectral information, resulting in optical simulated features. This step compensates for the absence of spectral information caused by cloud occlusion. Subsequently, in the CGCF module, the optical simulated features are used to selectively supplement the cloudy optical features, yielding optical corrected features. These optical corrected features are then fused with the SAR features to obtain the fused semantic features. This procedure is repeated four times, with the optical feature extractor in each subsequent layer using the optical corrected features from the previous layer. As a result, the cloud impact is progressively eliminated layer by layer, ultimately producing the corrected optical features $\{\mathbf{O}_i^C\}_{i=1}^4$ and fused semantic features $\{\mathbf{F}_i^{\text{Fuse}}\}_{i=1}^4$ where i denotes the ith feature extraction encoder. Finally, we design a multitask learning framework: 1) the optical corrected features are used for the cloud removal task and 2) the fused semantic features are used for the semantic segmentation task. By simultaneously conducting these two tasks, the network can optimize their interdependencies throughout training, eventually enhancing the accuracy of both tasks.

B. Frequency Difference Cloud Mapping

The Haar wavelet transform can decompose an image's spatial-domain features into low- and high-frequency domain components. The low-frequency components are obtained as an average of neighboring features, representing the overall brightness or energy distribution of a region, whereas the high-frequency components correspond to the gradient among neighboring features, capturing local variations, textures, and edge information [37]. For the low-frequency components, clouds in optical images generally appear as large-scale brightness increases and loss of fine details. Consequently, when optical features are cloud covered, the low-frequency energy increases overall, resulting in significant deviations from the corresponding SAR features. Meanwhile, for high-frequency components, the edges of clouds induce abrupt brightness transitions in the optical features, leading to pronounced high-frequency coefficients. As SAR features preserve ground object edges due to cloud penetration, the high-frequency differences are mainly concentrated along cloud boundaries, allowing precise delineation of cloud edges. Therefore, by comparing the low-frequency and high-frequency component differences between the optical features and SAR features, it is possible to determine the boundaries and intensity of cloud-affected regions in the optical features, thereby generating the cloud impact map.

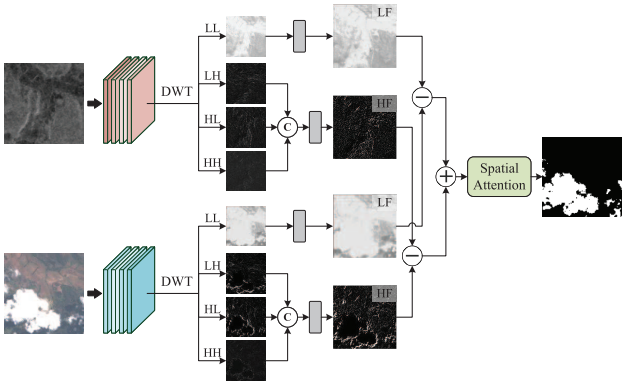


Fig. 3. Illustration of the FDCM. The gray rectangles represent the 1×1 convolutional operation and upsampling operation.

The structure of FDCM is illustrated in Fig. 3. $\mathbf{O}_i \in \mathbb{R}^{C_i \times H_i \times W_i}$ and $\mathbf{S}_i \in \mathbb{R}^{C_i \times H_i \times W_i}$ represent the optical features and SAR features extracted by the ith layer feature extractor, respectively. Here, C_i, H_i, and W_i denotes the channel dimensions, height, and width of the feature maps, respectively. We apply Haar wavelet transform to $\mathbf{O}_i$ and $\mathbf{S}_i$, starting with a 1-D decomposition along the row direction. The computation

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is performed as follows:

$$\begin{cases} \mathbf{A}_c(i,j) &= \frac{\mathbf{x}_c(2i-1,j)+\mathbf{x}_c(2i,j)}{2} \\ \mathbf{D}_c(i,j) &= \frac{\mathbf{x}_c(2i-1,j)-\mathbf{x}_c(2i,j)}{2} \end{cases} \tag{1}$$

where $\mathbf{x}_c \in \mathbb{R}^{\mathrm{H}_i \times W_i}$ represents the cth channel of $\mathbf{X}_c \in \mathbb{R}^{C_i \times \mathrm{H}_i \times W_i}$. $\mathbf{A}_c(i,j)$ and $\mathbf{D}_c(i,j)$ denote the row low-frequency coefficients and row high-frequency coefficients for channel c, respectively. i is the row index ranging from 1 to $H_i/2$, and j is the column index ranging from 1 to W_i. Then, the row low-frequency coefficients and row high-frequency coefficients are subjected to column decomposition. The computation is performed as follows:

$$\begin{cases} \mathbf{LL}_c(i,j) = \dfrac{\mathbf{A}_c(i,2j-1)+\mathbf{A}_c(i,2j)}{2} \\ \mathbf{HL}_c(i,j) = \dfrac{\mathbf{A}_c(i,2j-1)-\mathbf{A}_c(i,2j)}{2} \\ \mathbf{LH}_c(i,j) = \dfrac{\mathbf{D}_c(i,2j-1)+\mathbf{D}_c(i,2j)}{2} \\ \mathbf{HH}_c(i,j) = \dfrac{\mathbf{D}_c(i,2j-1)-\mathbf{D}_c(i,2j)}{2} \end{cases} \tag{2}$$

where i ranges from 1 to $H/2$ and j ranges from 1 to $W/2$. $\mathbf{LL}_c \in \mathbb{R}^{(\mathrm{H}_i/2)\times(W_i/2)}$, $\mathbf{HL}_c \in \mathbb{R}^{(\mathrm{H}_i/2)\times(W_i/2)}$, $\mathbf{LH}_c \in \mathbb{R}^{(\mathrm{H}_i/2)\times(W_i/2)}$, and $\mathbf{HH}_c \in \mathbb{R}^{(\mathrm{H}_i/2)\times(W_i/2)}$ represent the low-frequency coefficients, high-frequency horizontal coefficients, high-frequency vertical coefficients, and high-frequency diagonal coefficients for channel c, respectively. Then, we concatenated the three high-frequency components along the channel dimension and applied a 1×1 convolution for dimensionality reduction, which results in the final high-frequency and low-frequency features

$$\begin{cases} \mathbf{F}^{\mathrm{L}} = \mathrm{UP}\left(\delta_{1\times1}(\mathbf{LL})\right) \\ \mathbf{F}^{\mathrm{H}} = \mathrm{UP}\left(\delta_{1\times1}(\mathrm{Cat}(\mathbf{HL},\mathbf{LH},\mathbf{HH}))\right) \end{cases} \tag{3}$$

where $\mathbf{F}^{\mathrm{L}} \in \mathbb{R}^{C_i \times \mathrm{H}_i \times W_i}$ and $\mathbf{F}^{\mathrm{H}} \in \mathbb{R}^{C_i \times \mathrm{H}_i \times W_i}$ represent the low-frequency and high-frequency features, respectively. UP$(\cdot)$ denotes the upsampling operation, which restores the low-frequency and high-frequency features to the original feature size. $\delta_{1\times1}(\cdot)$ represents the 1×1 convolutional operation.

Let $\mathbf{O}_i^{\mathrm{L}}$ and $\mathbf{O}_i^{\mathrm{H}}$ and $\mathbf{S}_i^{\mathrm{L}}$ and $\mathbf{S}_i^{\mathrm{H}}$ be the low-frequency and high-frequency components of the optical and SAR features, respectively. We then subtract the low-frequency and high-frequency components of the SAR features from the corresponding components of the optical features, thereby obtaining the frequency differences between the optical and SAR features. After summing the low-frequency and high-frequency differences, we apply a spatial attention mechanism [38] to assign a cloud impact weight to each pixel. The spatial attention can adaptively assign higher weights to regions with greater frequency differences and lower weights to regions with smaller frequency differences, which aligns well with the objective of the FDCM. This process ultimately yields the cloud impact map for the entire image. The specific computation is as follows:

$$\hat{\mathbf{M}}_i = \mathrm{SA}\left(\alpha \times \left(\mathbf{O}_i^{\mathrm{L}} - \mathbf{S}_i^{\mathrm{L}}\right) + \beta \times \left(\mathbf{O}_i^{\mathrm{H}} - \mathbf{S}_i^{\mathrm{H}}\right)\right) \tag{4}$$

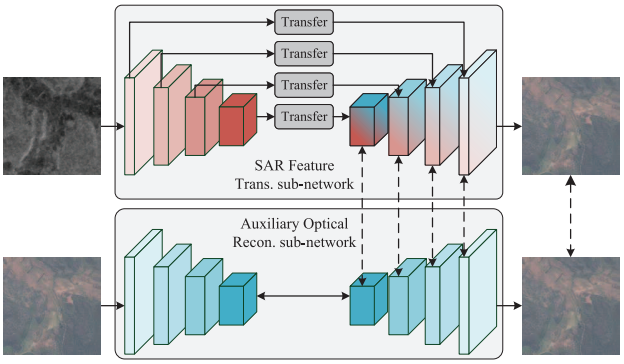


Fig. 4. Illustration of the SOFT. The dashed lines indicate that the two are constrained by a loss function.

where $\hat{\mathbf{M}}_i$ is the generated cloud impact map and α and β are the learnable parameters used to adaptively adjust the weight of the low-frequency and high-frequency differences in generating the cloud impact map. SA$(\cdot)$ represents the spatial attention mechanism.

C. SAR-to-OPT Feature Transfer

The purpose of the SOFT module is to use SAR scattering features to generate the corresponding optical spectral features that are missing due to cloud occlusion. Its structure is shown in Fig. 4, consisting of an auxiliary optical reconstruction subnetwork and an SAR feature transfer subnetwork. The former is used to guide the latter in generating more realistic optical spectral features. Both subnetworks follow an encoder–decoder architecture. In the auxiliary optical reconstruction subnetwork, the encoder progressively extracts features from the optical image, which contain rich spectral information, allowing the decoder to effectively reconstruct the original optical image. In the SAR feature transfer subnetwork, the encoder progressively extracts features from the SAR image, which contains scattering information. Between the encoder and the decoder, a feature transfer module is added to learn the mapping from scattering features to spectral features. The decoder then uses the optical simulated features to generate the translated optical image. The goal of the SAR feature transfer subnetwork is not to generate realistic optical images, but to promote the simulated features to approach the real optical features. This is achieved by applying a constraint that is based on the translated optical image and the original optical image.

In the training stage, a set of cloud-free optical images and SAR images from the identical region is combined. The optical image reconstruction task is performed first, using the cloud-free optical image to train the auxiliary optical reconstruction subnetwork. This network progressively extracts deep features from the optical image and reconstructs the original optical image through the decoder. Through training, the auxiliary optical reconstruction subnetwork can effectively extract the latent spectral features. Once the training of the auxiliary optical reconstruction subnetwork is completed, its parameters are frozen. Next, we use a set of cloud-free optical images and SAR images to train the SAR feature transfer subnetwork. In addition to use the original optical image as supervision, the trained auxiliary optical reconstruction subnetwork is also

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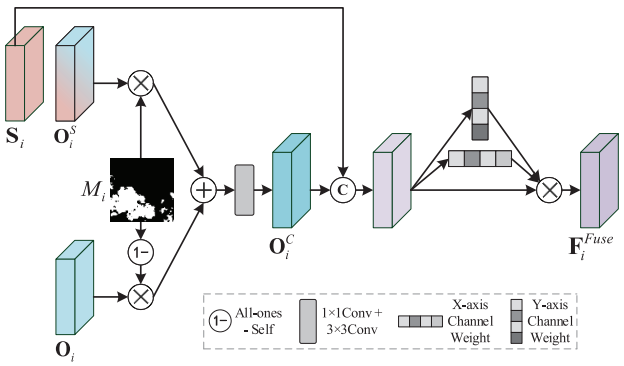


Fig. 5. Illustration of the CGCF.

used to extract spectral features from the optical image, which jointly guides the SAR feature transfer subnetwork to generate high-quality optical simulated features. The loss function is formulated as follows:

$$\text{Loss} = \frac{1}{H \times W} \sum_{i=1}^{\text{H}} \sum_{j=1}^{W} \left| I^{\text{cf}}_{(i,j)} - \hat{I}^{\text{cf}}_{(i,j)} \right| + \frac{1}{4} \sum_{k=1}^{4} \frac{1}{H_k \times W_k} \sum_{i=1}^{\text{H}_k} \sum_{j=1}^{W_k} \left(\mathbf{O}^R_{k,(i,j)} - \mathbf{O}^S_{k,(i,j)} \right)^2 \quad (5)$$

where I^{cf} represents the original cloud-free optical image, $\hat{I}^{\text{cf}}$ represents the optical image generated by the SAR feature transfer subnetwork, and $\mathbf{O}^R_k$ and $\mathbf{O}^S_k$ denote the kth layer decoder features of the auxiliary optical reconstruction subnetwork and the SAR feature transfer subnetwork, respectively. H_k and W_k represent the height and width of the kth layer decoder features, respectively. By minimizing the difference between the features in both subnetworks, the SAR feature transfer subnetwork is able to transfer the SAR features to the optical simulated features while generating the optical image.

It is notable that the training of the SOFT module is trained separately. After the training is completed, only the SAR feature transfer subnetwork is used to generate the optical simulated features. In the CGCF module, these features are then used to complement the cloud-affected optical features.

D. Cloud-Guided Complement Fusion

The CGCF module serves two purposes. First, it corrects the optical features affected by cloud interference. Second, it performs multimodal semantic fusion by combining the optical corrected features with the SAR features. The architecture of the CGCF module is shown in Fig. 5.

As shown in Fig. 5, we use the cloud impact map as a reference. For regions in optical features that are affected by clouds, we prioritize the use of the optical simulated features for supplementation. In contrast, for regions with minimal cloud impact, we retain the original optical features as much as possible. The specific computational process is as follows:

$$\mathbf{O}^C_i = \text{Conv}\left(\mathbf{O}^S_i \times \hat{\mathbf{M}}_i + \mathbf{O}_i \times \left(\mathbf{1} - \hat{\mathbf{M}}_i \right) \right) \quad (6)$$

where Conv(·) represents a convolutional block containing 1×1 and 3×3 kernels, $\mathbf{1}$ denotes an all-ones matrix of the same size as $\hat{\mathbf{M}}_i$, $\mathbf{O}^C_i$ represents the optical corrected features, $\mathbf{O}^S_i$ denotes the optical simulated features, and $\mathbf{O}_i$ represents the original optical features. Intuitively, $\hat{\mathbf{M}}_i$ indicate the ratio of optical simulated features to original optical features. In $\hat{\mathbf{M}}_i$, larger values indicate regions significantly affected by clouds, where the original optical features are less valuable, thus require supplementation by the optical simulated features. Smaller values in $\hat{\mathbf{M}}_i$ indicate regions with minimal cloud impact, where the original optical features provide a more accurate representation and therefore should be retained more.

Next, we use the optical corrected features and the original SAR features for multimodal semantic fusion to achieve a more comprehensive semantic representation. The optical corrected features and SAR features are combined by concatenating them along the channel dimension. Since the features from the corrected optical branch and the SAR branch may exhibit varying degrees of importance, it is necessary to assign different weights to the features of these two branches. Compared with the channel attention [38], coordinate attention [39] can simultaneously capture interchannel dependencies as well as spatial positional information. Consequently, when performing feature aggregation under potential cloud interference, it is better to apply coordinate attention than channel attention to consider both the channel and spatial influences. The specific computational process is as follows:

$$\mathbf{F}^{\text{Fuse}}_i = \text{CoordA}\left(\text{Cat}\left(\mathbf{O}^C_i, \mathbf{S}_i \right) \right) \quad (7)$$

where CoordA(·) represents the coordinate attention mechanism, Cat(·) denotes the channel concatenation operation, and $\mathbf{F}^{\text{Fuse}}_i$ represents the multimodal fused semantic features.

After the above operations, we obtain the optical corrected features and the fused semantic features. These are then passed through the subsequent cloud removal decoder and semantic segmentation decoder, respectively, to obtain the cloud-removed optical images and the land cover maps.

E. Multitask Decoder and Loss Function

The decoders used in both the semantic segmentation and cloud removal tasks are based on the U-Net decoder architecture. Low-resolution features usually encompass more semantic information, while high-resolution features capture more structural details. Therefore, the decoder utilizes multiscale feature fusion to perform semantic segmentation and cloud removal. In the semantic segmentation task, since the fused semantic features $\{\mathbf{F}^{\text{Fuse}}_i\}^4_{i=1}$ integrate information from both optical and SAR features, the semantic expression becomes more comprehensive. As a result, we use the fused semantic features and the cross-entropy loss function for the semantic segmentation task. The formula is shown as follows:

$$L_{\text{seg}} = -\frac{1}{H \times W} \sum_{c=1}^{C} \sum_{i=1}^{\text{H}} \sum_{j=1}^{W} y_{c,(i,j)} \log\left(\hat{y}_{c,(i,j)} \right) \quad (8)$$

where C denotes the number of land cover classes and $y_{c,(i,j)}$ and $\hat{y}_{c,(i,j)}$ represent the true labels and predicted probabilities for class c at position (i, j), respectively.

In the cloud removal task, the objective is to restore the cloud-affected areas of the optical image. Since the optical corrected features $\{\mathbf{O}^C_i\}^4_{i=1}$ successfully use optical simulated features to supplement the cloud-affected optical features, we

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adopt the optical corrected features to generate the cloud-removed image. For the cloud removal task, the focus is on reconstructing the ground conditions in the cloud-affected regions. Therefore, we make an additional constraint on the cloud-affected areas

$$L_{\text{rec}} = \frac{1}{H \times W} \sum_{i=1}^{\text{H}} \sum_{j=1}^{W} \left(1 + \lambda \mathbf{M}_{(i,j)}\right) \times \left| I^{\text{cf}}_{(i,j)} - I^{\text{cr}}_{(i,j)} \right| \tag{9}$$

where I^{cf} and I^{cr} correspond to the true cloud-free image and the cloud-removed image, respectively. $\mathbf{M}$ indicates the cloud coverage condition of the pixels, where $\mathbf{M}_{(i,j)} = 1$ presents the pixels at position (i, j) that are covered by clouds, and $\mathbf{M}_{(i,j)} = 0$ represents that the pixels are not covered by clouds. λ denotes the additional constraint applied to the cloud-covered regions. When $\lambda = 0$ is true, L_{rec} degenerates into mean absolute error (MAE) loss function.

The overall loss function is formulated as follows:

$$L_{\text{total}} = \alpha \times L_{\text{seg}} + \beta \times L_{\text{rec}} \tag{10}$$

where α and β are the balancing parameters.

IV. Experiments

A. Experimental Settings

1) Datasets: We train and benchmark the proposed SegCR on the M3M-CR dataset [15] as well as the LuojiaSET-OSFCR dataset [40]. *The M3M-CR dataset* contains 63 000 optical–SAR image pairs in total. The optical images are obtained from the PlanetScope Level-3B product at an altitude of approximately 475 km. These images have a spatial resolution of 3 m and a size of 300 pixels and cover four spectral bands (blue, green, red, and near infrared). The SAR images are sourced from ESA Sentinel-1 A/B satellites at an altitude of about 693 km and IW mode with VV and VH polarization channels in the C band. These images have a spatial resolution of 10 m and a size of 90 pixels. To ensure temporal alignment, the average time interval between SAR and optical acquisitions is around 2.2 days, while the mean interval between cloudy and cloud-free optical pairs is about 2.8 days. In the experiments, only the RGB bands of the optical images are used, and the sizes of both the optical and SAR images are unified to 256 pixels using bilinear interpolation.

The LuojiaSET-OSFCR dataset consists of 20 000 optical–SAR image pairs. The optical imagery is sourced from the Sentinel-2 Level-1C top-of-atmosphere reflectance product, including 13 spectral bands covering visible to near-infrared wavelengths. The original bands, with spatial resolutions of 10, 20, and 60 m, were resampled to a unified resolution of 10 m. The SAR imagery is derived from Sentinel-1 Level-1 GRD data acquired in the C band and IW mode with VV and VH dual polarizations. Both modalities are co-registered to ensure spatial consistency with a size of 256 pixels. In the experiments, to maintain consistency with the M3M-CR dataset, only the RGB bands of the optical images are used.

2) Evaluation Metrics: We assess the proposed SegCR along with other comparative approaches using frequently adopted evaluation metrics in the domains of semantic segmentation and cloud removal.

a) Semantic segmentation: We employ two commonly adopted quantitative assessment criteria to assess the semantic segmentation performance of our method as well as all comparative methods: mean intersection over union (mIoU) and mean pixel accuracy (mPA). The mIoU measures the extent to which the regions predicted by a model coincide with the corresponding regions in the ground truth. Meanwhile, the mPA represents the proportion of pixels that are accurately classified for each individual class compared with the total number of pixels within that class. The expressions of mIoU and mPA are presented as follows:

$$\begin{aligned} \text{mIoU} &= \frac{1}{C} \sum_{c=1}^{C} \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c} \\ \text{mPA} &= \frac{1}{C} \sum_{c=1}^{C} \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c} \end{aligned} \tag{11}$$

where C represents the total number of land cover classes and TP_c, FP_c, and FN_c indicate the number of true positive, false positive, false negative pixels for class c, respectively. Greater values of these two metrics signify superior performance in semantic segmentation.

b) Cloud removal: To assess our method and all comparative methods in restoring cloud-covered regions, we use three widely used image quality assessment criteria, including peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM), and MAE.

PSNR evaluates image quality by computing the ratio of the maximum achievable signal strength to the noise level. The calculation procedure is shown as follows:

$$\text{PSNR}\,(x, y) = 10 \log \left(\frac{\text{MAX}^2}{\text{mse}} \right) \tag{12}$$

where MAX represents the maximum pixel value in the generated image x and the ground-truth image y and mse denotes the mean squared error between the generated image and the ground-truth image. A larger PSNR value implies that the quality of the generated image is better.

SSIM measures image quality by comparing the similarity in three major aspects: luminance, contrast, and structure, which are more in line with human visual perception. The calculation procedure is shown as follows:

$$\text{SSIM}\,(x, y) = \frac{\left(2\mu_x\mu_y + c_1\right)\left(2\sigma_{xy} + c_2\right)}{\left(\mu_x^2 + \mu_y^2 + c_1\right)\left(\sigma_x^2 + \sigma_y^2 + c_2\right)} \tag{13}$$

where μ_x and σ_x^2 represent the mean and variance of the generated image x, μ_y and σ_y^2 represent the mean and variance of the ground truth image y, μ_{xy} denotes the covariance between the generated image x and the ground-truth image y, and c_1 and c_2 are the two constants used to avoid division by 0. A larger SSIM value indicates better quality of the synthesized image.

MAE measures the similarity between two images by calculating the average of the absolute differences between

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TABLE I

QUANTITATIVE COMPARISONS BETWEEN THE PROPOSED SEGCR AND THE CURRENT SOTA METHODS IN THE FIELD OF LAND COVER MAPPING. THE RESULTS WITH THE HIGHEST PERFORMANCE ARE SHOWN IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Method	M3M-CR						LuojiaSET-OSFCR					
	Cloudy		Cloud-Free		Overall		Cloudy		Cloud-Free		Overall	
	mPA	mIoU	mPA	mIoU	mPA	mIoU	mPA	mIoU	mPA	mIoU	mPA	mIoU
MCANet [41]	53.38	39.24	63.68	52.09	59.58	46.28	57.69	43.60	67.08	56.41	61.94	49.96
LASNet [42]	54.57	41.30	65.13	53.87	61.25	47.62	59.27	45.22	68.97	57.39	63.15	51.06
PACSCNet [43]	55.80	42.57	66.34	54.26	61.88	48.68	58.08	44.14	70.58	58.45	64.29	51.13
SoftFormer [44]	58.74	45.92	67.87	55.97	62.66	50.12	60.93	46.69	70.66	58.12	64.94	52.42
FTransUNet [25]	60.85	46.23	67.44	54.80	62.47	50.37	62.15	48.89	72.68	59.01	66.49	53.78
CMX [45]	60.39	47.09	67.12	55.78	63.32	51.46	61.09	48.77	**74.23**	**60.62**	66.82	53.36
CloudSeg [16]	62.96	50.28	67.09	55.04	65.74	52.69	65.48	54.59	73.06	59.26	70.87	57.14
SegCR (ResNet50)	64.82	51.97	68.34	55.66	66.56	53.48	69.07	55.42	72.87	59.13	71.11	57.65
SegCR (Ours)	**65.78**	**52.30**	**69.74**	**56.23**	**67.82**	**54.75**	**70.12**	**56.41**	73.32	59.96	**72.04**	**58.11**

corresponding pixel values. The calculation procedure is shown as follows:

$$\mathrm{MAE} = \frac{1}{H \times W} \sum_{i=1}^{\mathrm{H}} \sum_{j=1}^{W} \left| x_{i,j} - y_{i,j} \right| \tag{14}$$

where $x_{i,j}$ and $y_{i,j}$ denote the pixel values in the ith row and the jth column of the generated image x and the ground-truth image y, respectively. A smaller MAE value indicates better quality of the generated image.

3) Implementation Details: We use a 7:1:2 ratio to divide each dataset into training, validation, and testing sets [45]. The proposed SegCR framework is trained and evaluated using an NVIDIA RTX 3090 24G GPU on the Pytorch architecture. First, we trained the SOFT module. For data augmentation, we applied random rotations and flips. The AdamW optimizer was used with an initial learning rate of $1e^{-3}$ and a weight decay of $5e^{-4}$. The batch size was set to 1, and the maximum number of training epochs was set to 50. After completing the training of the SOFT module, we froze its parameters and proceeded to train the remaining part of SegCR. Similar data augmentation techniques with the process of the SOFT module were applied, and the AdamW optimizer was used again with an initial learning rate of $1e^{-3}$ and a weight decay of $5e^{-4}$. A cosine annealing learning rate decay strategy was adopted, reducing the learning rate to a minimum value of $5e^{-4}$ in the final training epochs. The batch size was set to 16, and the maximum number of training epochs was set to 100. The parameters of the feature extractor were initialized with the pretrained Segformer-B4 model, while the parameters of other newly added convolutional layers were initialized using the Kaiming method. The value of λ in L_{rec} was set to 5, and the weighting factors α and β were set to 1 empirically.

B. Comparison With Semantic Segmentation Methods

We benchmarked the proposed SegCR and compared with SOTA multimodal semantic segmentation methods, including FTransUNet [25], SoftFormer [44], CMX [45], LASNet [42], MCANet [41], Cloudseg [16], and PACSCNet [43]. In addition to evaluating the overall performance of SegCR and all comparison methods in two datasets, we also divided the datasets into cloudy and cloud-free sections based on cloud coverage maps and calculated accuracy for each section separately to assess the method's specificity under cloudy conditions and stability under cloud-free conditions. Table I shows the quantitative results. From the results, it can be observed that the proposed SegCR outperforms the current SOTA methods in overall performance on both benchmark datasets and demonstrates strong competitiveness in both cloudy and cloud-free scenarios. Specifically, on the M3M-CR dataset, our method achieves 54.75% overall mIoU and 67.82% overall mPA, outperforming the best-performing Cloudseg by 2.06% and 2.08%, respectively. Regarding the LuojiaSET-OSFCR dataset, our method achieves 58.11% overall mIoU and 72.04% overall mPA, surpassing Cloudseg by 0.97% and 1.17%, respectively. In cloud-free scenarios, our method achieves comparable accuracy with other approaches, as the absence of cloud occlusion allows optical spectral information to be fully captured and effectively fused with SAR scattering information. However, in cloudy scenarios, our method exhibits a significant advantage over others. This is because cloud occlusion severely degrades optical spectral information, rendering it ineffective and causing conflicts with SAR scattering information, which confuses conventional models when interpreting land cover in cloud-covered regions. In contrast, our proposed SAR feature transfer subnetwork effectively leverages SAR scattering information to reconstruct the missing spectral information, thus mitigating the risk of model confusion and enhancing segmentation performance in cloud-covered areas.

We selected four representative scenes from the M3M-CR and LuojiaSET-OSFCR datasets for qualitative analysis of semantic segmentation results, as shown in Fig. 6. Based on the visual outcomes, it is clearly observable that the land cover maps produced by the proposed SegCR exhibit higher consistency with the ground truth. For instance, in the first scene of M3M-CR dataset, in regions not covered by clouds, almost all methods achieve relatively accurate land cover mapping. However, in the lower part of the image, where a small

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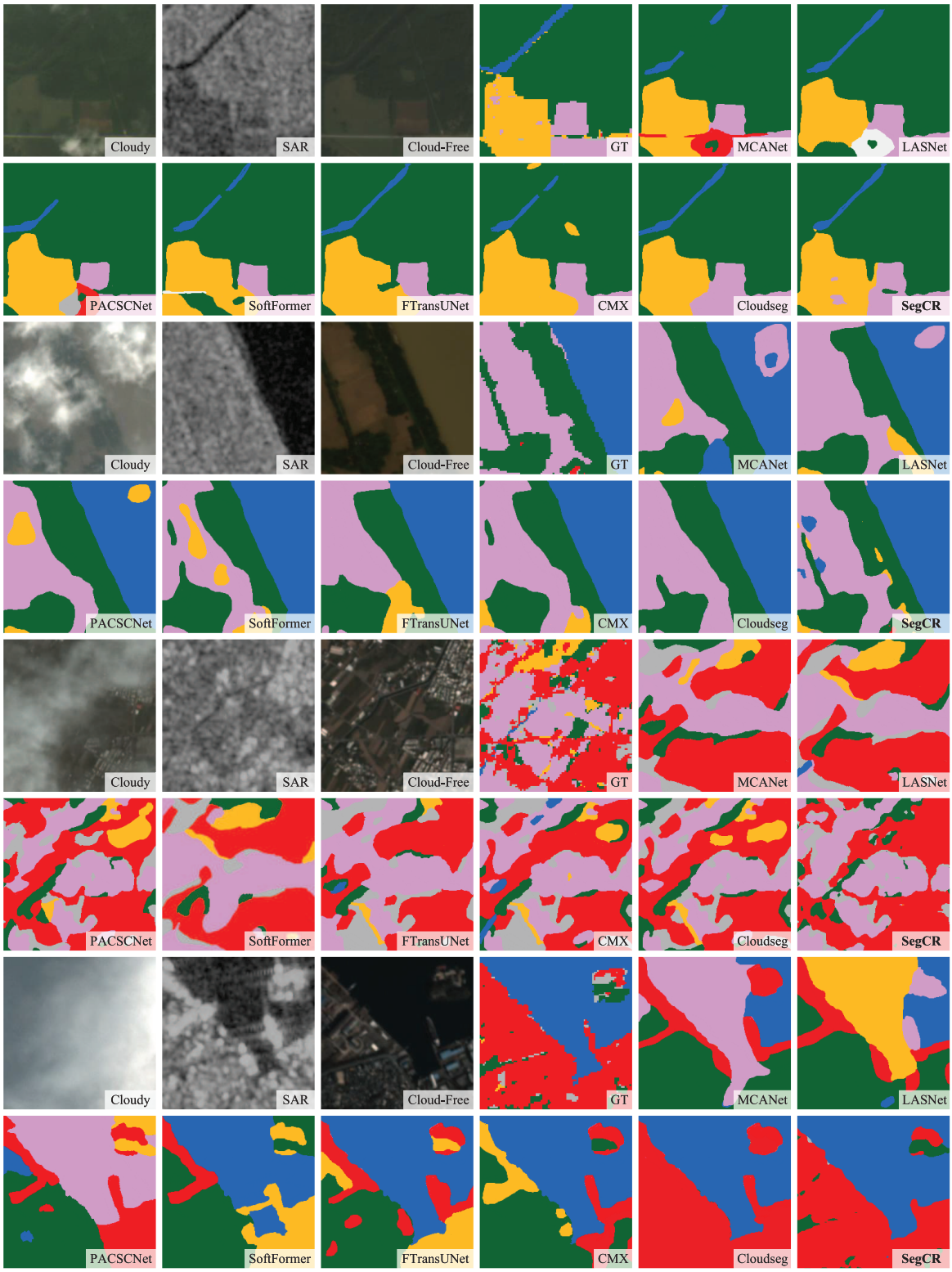


Fig. 6. Qualitative results of land cover mapping for four different scenes. The first two scenes are from the M3M-CR dataset, while the latter two are from the LuojiaSET-OSFCR dataset. For each scene, from top left to bottom right are, respectively, the cloudy image, the SAR image, the cloud-free image, the ground truth, the result from MCANet, LASNet, PACSCNet, SoftFormer, FTransUNet, CMX, Cloudseg, and our SegCR.

cloud-covered area exists, SegCR accurately distinguishes between grassland and agricultural land, whereas other methods fail to delineate their boundaries precisely. Furthermore, in the second scene of LuojiaSET-OSFCR dataset, where the optical image is almost entirely obscured by clouds, other comparative methods almost completely lose their classification capability. This failure occurs because these methods rely on consistent optical spectral and SAR scattering features for effective fusion. However, in this case, the severe inconsistency between the optical and SAR images leads to model failure.

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TABLE II

QUANTITATIVE COMPARISONS BETWEEN THE PROPOSED SEGCR AND THE CURRENT SOTA METHODS IN THE FIELD OF CLOUD REMOVAL. THE RESULTS WITH THE HIGHEST PERFORMANCE ARE SHOWN IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Method	M3M-CR			LuojiaSET-OSFCR		
	SSIM↑	PSNR↑ (dB)	MAE↓ (10^{-2})	SSIM↑	PSNR↑ (dB)	MAE↓ (10^{-2})
HS2P [29]	0.8789	28.69	2.95	0.8573	28.24	3.04
USSRN-CR [28]	0.8842	29.78	2.74	0.8659	29.48	2.83
CloudSeg [16]	0.8846	29.44	2.72	0.8678	29.44	2.77
Align-CR [15]	0.8874	30.06	2.66	0.8786	30.12	2.72
GLF-CR [30]	0.9026	30.12	2.67	0.8794	29.63	2.75
SCT-CR [31]	<u>0.9064</u>	<u>30.26</u>	<u>2.47</u>	<u>0.8872</u>	<u>30.58</u>	<u>2.64</u>
SegCR (ResNet50)	0.9112	30.65	2.38	0.8894	30.52	2.60
SegCR (Ours)	**0.9150**	**31.24**	**2.23**	**0.8966**	**30.70**	**2.43**

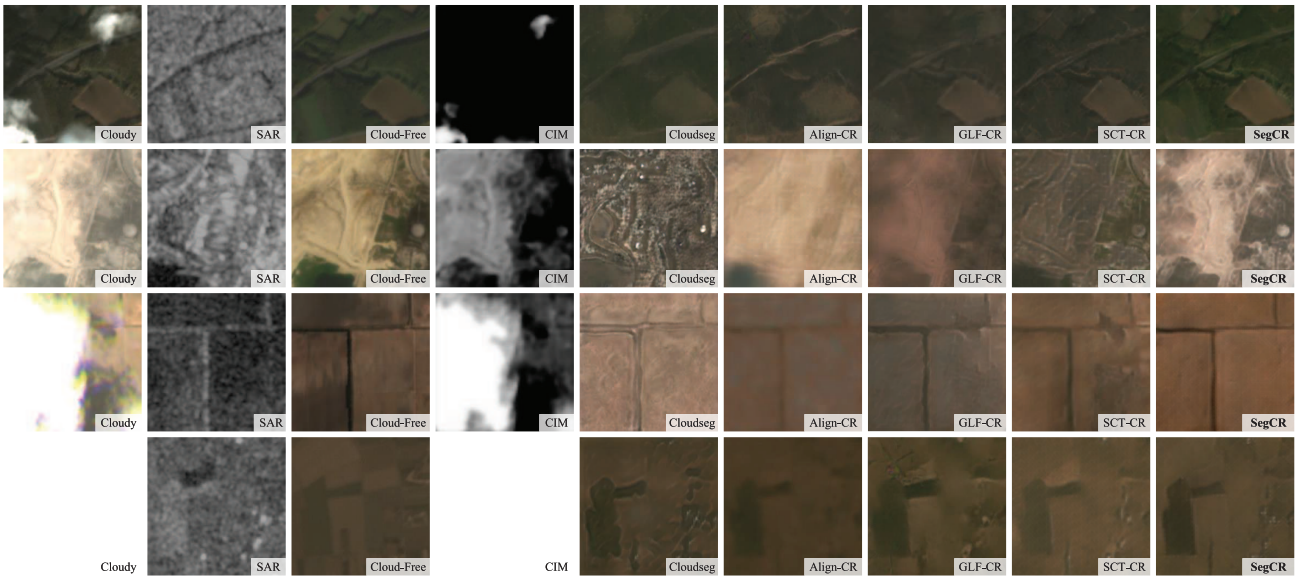


Fig. 7. Qualitative results of cloud removal for four different scenes. The first two scenes are from the M3M-CR dataset, while the latter two are from the LuojiaSET-OSFCR dataset. For each scene, from top left to bottom right are, respectively, the cloudy image, the cloud-free image, the SAR image, the cloud impact map generated by FDCM, the result from CloudSeg, Align-CR, GLF-CR, SCT-CR, and our SegCR.

In contrast, our method leverages the prior knowledge learned by the SOFT module to transfer SAR scattering features into the optical spectral domain, effectively compensating for the missing optical spectral features. Moreover, the CGCF module further facilitates multimodal feature fusion, enabling our method to achieve more accurate semantic segmentation under heavy cloud cover conditions.

C. Comparison With Cloud Removal Methods

We compared the proposed SegCR with SOTA multimodal cloud removal methods, including Align-CR [15], HS2P [29], SCT-CR [31], GLF-CR [30], Cloudseg [16], and USSRN-CR [28]. We evaluated the model's ability to reconstruct cloud-covered regions by comparing the overall cloud removal metrics on two datasets. The quantitative results are presented in Table II. As shown, SegCR outperforms current SOTA methods in cloud removal effectiveness on both benchmark datasets. Specifically, on the M3M-CR dataset, our method achieves an overall SSIM of 0.9150 and an overall PSNR of 31.24 dB, surpassing the best-performing SCT-CR by 0.86% and 0.98 dB, respectively. Regarding the LuojiaSET-OSFCR dataset, our method achieves an overall SSIM of 0.8966 and an overall PSNR of 30.70 dB, outperforming Align-CR by 0.94% and 0.12 dB. These results demonstrate that cloud-removed optical images generated by SegCR exhibit superior visual perception quality while minimizing pixel-level reconstruction errors, making it the most effective approach currently available for multimodal cloud removal.

We selected four representative scenes from the M3M-CR and LuojiaSET-OSFCR datasets to visualize cloud removal results for qualitative analysis, as shown in Fig. 7. From a visual perspective, the proposed SegCR demonstrates superior ability in reconstructing cloud-covered regions and restoring surface features. In addition, the FDCM module accurately locates cloud positions and assesses their impact, thereby

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providing guidance for the complementary fusion in the CGCF module. For example, in the first scene of the M3M-CR dataset, where cloud coverage is relatively sparse, SegCR preserves the original appearance of cloud-free regions more effectively than other methods which tend to degrade these areas. This is attributed to the FDCM module, which accurately identifies regions affected by cloud cover. Consequently, in the subsequent CGCF module, only cloud-affected optical features are supplemented and corrected, while unaffected optical features remain intact. In addition, in the second scene of the LuojiaSET-OSFCR dataset, where the optical image is entirely obscured by clouds, SegCR still reconstructs surface features effectively, whereas other methods perform poorly. This advantage arises from the proposed SOFT module, which, through the pretraining phase, learns the prior knowledge of transferring SAR scattering features into optical spectral features. As a result, even when optical spectral information is completely missing, SegCR can generate realistic optical simulated features from SAR scattering features to facilitate cloud-covered optical image reconstruction. In contrast, because of the lack of crucial optical spectral information, other methods are unable to restore the cloud-covered regions efficiently.

D. Ablation Study

The proposed SegCR first utilizes the SegFormer-B2 backbone to extract deep feature of the optical–SAR image pairs. Then, the FDCM module extracts the cloud impact map, which serves as guidance for subsequent multimodal complementary fusion. Next, the SOFT module transfers SAR scattering features into optical simulated features, compensating for the optical spectral information lost due to cloud occlusion. Then, in the CGCF module, based on the cloud impact map, SAR features, optical simulated features, and original optical features are selectively fused to obtain optical corrected features and fused semantic features. Finally, we establish a multitask decoder architecture that can generate land cover maps and cloud-removed optical images concurrently. By means of the reciprocal cooperation between the high-level semantic comprehension task and the low-level vision task, both the performance of cloud removal and the precision of semantic segmentation are enhanced. To determine the contributions of each component to the performance of SegCR, we conducted comparative experiments and performance analysis by modifying or removing key modules of SegCR. Sections IV-D1–IV-D4 will detail the implementation of the ablation study.

1) Contributions of SegFormer-B2 Backbone: To determine whether the performance gains originate mainly from the strong SegFormer backbone or from the proposed strategy itself, we replaced the SegFormer backbone with a weaker feature extractor ResNet50. The experimental results are presented in the "SegCR (ResNet50)" row in Tables I and II. In addition, we conducted an ablation experiment by removing all proposed modules and retaining only the optical–SAR dual-stream SegFormer backbone. To achieve multimodal feature aggregation, we adopt a simple channel concatenation strategy to merge the optical and SAR features as the fused feature.

TABLE III
QUANTITATIVE OUTCOMES FOR EVALUATING THE EFFECTIVENESS OF EVERY COMPONENT OF PROPOSED SEGCR IN LAND COVER MAPPING

Method	M3M-CR		LuojiaSET-OSFCR	
	mPA	mIoU	mPA	mIoU
w/o FDCM	65.97	52.81	69.62	56.48
w/o SOFT	62.13	49.55	66.38	53.46
w/o Cloud Removal	65.32	52.66	70.36	57.07
SegFormer Baseline	61.43	48.72	63.87	51.75
SegCR (Ours)	**67.82**	**54.75**	**72.04**	**58.11**

TABLE IV
QUANTITATIVE OUTCOMES FOR EVALUATING THE EFFECTIVENESS OF EVERY COMPONENT OF PROPOSED SEGCR IN CLOUD REMOVAL

Method	M3M-CR		LuojiaSET-OSFCR	
	SSIM	PSNR	SSIM	PSNR
w/o FDCM	0.8968	30.12	0.8792	29.34
w/o SOFT	0.8646	28.26	0.8583	28.47
w/o Semantic Segmentation	0.8946	29.66	0.8726	29.26
SegFormer Baseline	0.8504	27.35	0.8447	27.12
SegCR (Ours)	**0.9150**	**31.24**	**0.8966**	**30.70**

The results are presented in the *SegFormer Baseline* row of Tables III and IV. Similar to the SegCR using SegFormer-B2 as the backbone, the SegCR using ResNet50 as the backbone significantly outperforms the comparison methods. In addition, whether the FDCM module, the SOFT module, or the auxiliary task branch is incorporated, the network performance consistently improves over the baseline. This indicates that the powerful backbone, indeed, contributes to some improvement in network performance, but the majority of the performance gain is primarily attributed to the proposed modules.

2) Effectiveness of FDCM: To facilitate a better understanding of the workflow of the FDCM module, we have visualized the intermediate results produced by FDCM. Fig. 8 demonstrates that the frequency components of SAR and cloud-free optical regions are consistent, while significant discrepancies arise in the cloud-covered areas. In addition, the low-frequency difference map between the cloud-contaminated optical image and the SAR image can roughly identify the coverage area and intensity of the cloud layer, while the high-frequency difference map effectively delineates the boundaries of the cloud-covered regions. The cloud impact map, derived by weighted fusion of frequency differences followed by a spatial attention mechanism, accurately highlights regions affected by cloud occlusion.

To verify the contribution of the FDCM module to the proposed SegCR, we remove the FDCM module and retrain the SegCR. Without the cloud impact map extracted by FDCM as guidance, in the CGCF module, we directly added the optical simulated features to the original optical features, while keeping all other components unchanged. The experimental results for the model without the FDCM module are represented as the *w/o FDCM* row in Tables III and IV. Compared

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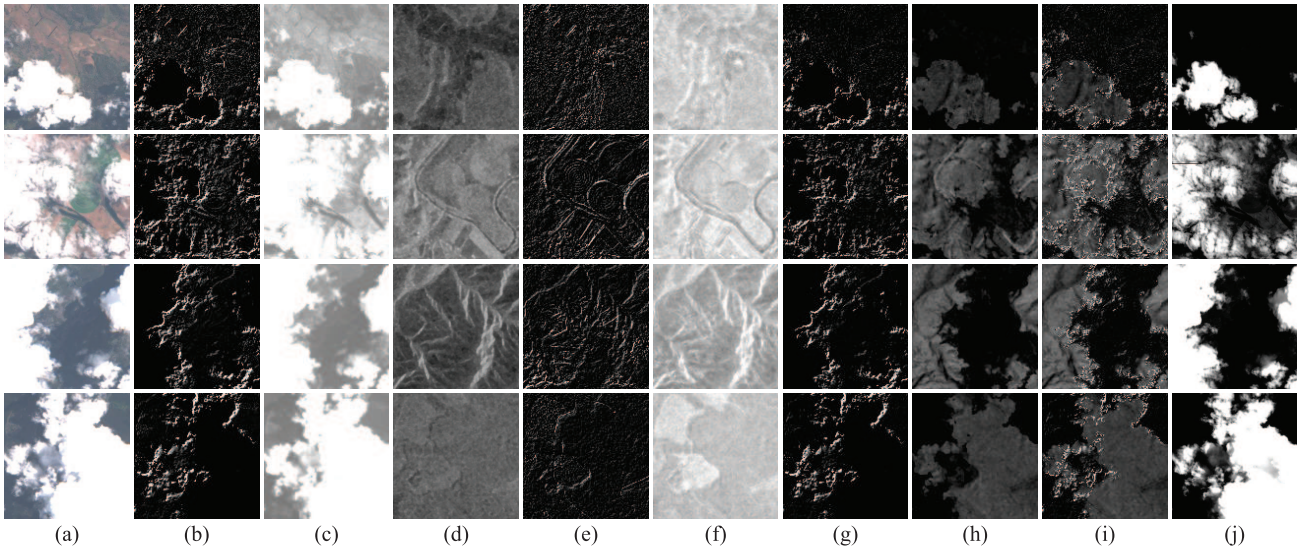


Fig. 8. (Left to Right) Intermediate results of the FDCM module. (a) Cloudy optical image. (b) High-frequency components of the cloudy optical image. (c) Low-frequency components of the cloudy optical image. (d) SAR image. (e) High-frequency components of the SAR image. (f) Low-frequency components of the SAR image. (g) High-frequency difference maps between the optical and SAR images. (h) Low-frequency difference maps between the optical and SAR images. (i) Sum of the high- and low-frequency difference maps. (j) Resulting cloud impact map.

with the complete SegCR network, both semantic segmentation and cloud removal performance exhibit a slight decline. This degradation occurs despite our efforts to generate realistic optical simulated features using the SOFT module. The key issue is that there remains a discrepancy between the optical simulated features and the original optical features. Without selective correction based on the cloud impact map, the model indiscriminately applies the optical simulated features to all regions of the optical image. Consequently, valuable regions that are not affected by cloud coverage may be unintentionally altered, leading to decreased performance in both semantic segmentation and cloud removal.

3) Effectiveness of SOFT: To validate the contribution of the SOFT module, we conducted an ablation study by removing it from SegCR. Without the SOFT module, the model could no longer generate optical simulated features using SAR scattering features. As a result, in the CGCF module, we replaced the optical simulated features with the original SAR features, which were then fused with the original optical features to generate optical corrected features, while keeping all other components unchanged. The experimental results for the model without the SOFT module are represented as the *w/o SOFT* row in Tables III and IV. Compared with the complete SegCR framework, both semantic segmentation performance and cloud removal performance exhibit a significant decline. This degradation is primarily due to the absence of optical simulated features derived from SAR scattering features, which are crucial for compensating for the missing optical spectral information caused by cloud occlusion. Furthermore, there exists a substantial semantic gap between SAR scattering features and optical spectral features. Directly using SAR scattering features to complement original optical features may lead to suboptimal fusion results, thereby negatively impacting both cloud removal and semantic segmentation accuracy.

4) Effectiveness of the Multitask Learning Framework: To validate the mutual benefits of the multitask learning framework for cloud removal and semantic segmentation, we conducted ablation experiments by separately removing the cloud removal decoder branch and the semantic segmentation decoder branch. The experimental results are represented as the *w/o Decloud* row in Table III and the *w/o semantic segmentation* row in Table IV, respectively. From the results, it is clear that the removal of any one of the branches causes a decrease in the performance of the remaining task. This can be attributed to the introduction of the cloud removal branch, which contributes to the reduction of semantic uncertainties in the areas affected by clouds. As a result, it leads to an improvement in the outcome of semantic segmentation. Conversely, the introduction of the semantic segmentation branch enhances the generation of high quality, semantically enriched optical corrected features, which are crucial for cloud removal. These experiments demonstrate that jointly performing the high-level semantic comprehensive task and low-level vision task enables both cloud removal and semantic segmentation to achieve the optimal performance.

In addition, we conducted comparative experiments with different combinations of α and β to find appropriate multitask loss balancing parameters. As shown in Table V, varying the weights within a reasonable range did not yield significant

TABLE V
QUANTITATIVE OUTCOMES FOR EVALUATING DIFFERENT MULTITASK LOSS BALANCING PARAMETERS ON SEGCR PERFORMANCE IN LAND COVER MAPPING AND CLOUD REMOVAL

Ratio of multitask loss	M3M-CR		LuojiaSET-OSFCR	
	mIoU	SSIM	mIoU	SSIM
$\alpha = 0.5, \beta = 1.5$	54.54	0.9153	57.98	0.8954
$\alpha = 1, \beta = 1$	54.75	0.9150	58.11	0.8966
$\alpha = 1.5, \beta = 0.5$	54.76	0.9132	58.13	0.8939

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performance improvement. Specifically, assigning $\alpha = \beta = 1$ achieves a balanced optimization between cloud removal and semantic segmentation, ensuring stable convergence without sacrificing accuracy in either task. These results empirically justify our choice of equal weights in the loss function.

V. Conclusion

In this article, we propose a multimodal and multitask complementary fusion network named SegCR to address the performance degradation caused by cloud interference in optical–SAR fusion for land cover mapping. First, we propose an FDCM module to extract the cloud impact map that guides the subsequent optical correction process by selectively preserving cloud-free optical features while correcting cloud-affected optical features. Then, we propose a SOFT module which leverages an optical reconstruction subtask and an SAR-to-OPT translation subtask to transfer SAR scattering features to optical simulated features, addressing the issue of optical information loss caused by cloud occlusion. Next, in the CGCF module, the cloud-affected optical features are corrected based on the cloud impact map, and the optical corrected features are fused with SAR features to obtain multimodal semantic fusion features with richer semantic information. Extensive comparative experiments on the M3M-CR and LuojiaSET-OSFCR datasets showcase the superiority of the proposed SegCR network. Furthermore, a sequence of ablation studies validate the effectiveness of every module within SegCR.

In upcoming research endeavors, our aim is to implement SegCR in real-life situations. Although our method has achieved promising results on laboratory datasets, real-world applications involve additional challenges, such as variations in illumination, imaging quality, and modality gaps. We will further refine our proposed approach to address these complexities and enhance its robustness in practical settings.

References

[1] G. Zhou, W. Chen, X. Qin, J. Li, and L. Wang, "Lithological unit classification based on geological knowledge-guided deep learning framework for optical stereo mapping satellite imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5624916.

[2] Y. Yuan, F. Li, D. Zhou, L. Bai, A. Jurek-Loughrey, and Z. Wang, "Extraction of aquaculture cages from high-resolution remote sensing images based on deep learning," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2024, pp. 9556–9560.

[3] Y. Du, Q. Sheng, W. Zhang, C. Zhu, J. Li, and B. Wang, "From local context-aware to non-local: A road extraction network via guidance of multi-spectral image," *ISPRS J. Photogramm. Remote Sens.*, vol. 203, pp. 230–245, Sep. 2023.

[4] Y. Fu, W. Li, S. Fan, Y. Jiang, and H. Bai, "CAL-Net: Conditional attention lightweight network for in-orbit landslide detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4408515.

[5] W. Bao, W. Feng, and R. Ma, "Remote sensing image classification based on support vector machine with the multi-scale segmentation," *Proc. SPIE*, vol. 9817, pp. 312–318, Dec. 2015.

[6] W. Feng, H. Sui, W. Huang, C. Xu, and K. An, "Water body extraction from very high-resolution remote sensing imagery using deep U-Net and a superpixel-based conditional random field model," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 4, pp. 618–622, Apr. 2019.

[7] W. Miao, J. Geng, and W. Jiang, "Multigranularity decoupling network with pseudolabel selection for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5603813.

[8] K. Mansour et al., "Phytoplankton impact on marine cloud microphysical properties over the northeast Atlantic Ocean," *J. Geophys. Res., Atmos.*, vol. 127, no. 10, p. 2021, May 2022.

[9] D. Wei, H. Xie, P. Li, and Y. Xu, "A learning framework with multispectral band-differentiated encoding for remote sensing water body detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 6278–6289, 2024.

[10] Z. Xu, W. Jiang, and J. Geng, "Dual-branch dynamic modulation network for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514813.

[11] J. Geng, X. Deng, X. Ma, and W. Jiang, "Transfer learning for SAR image classification via deep joint distribution adaptation networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5377–5392, Aug. 2020.

[12] R. Bamler, "Principles of synthetic aperture radar," *Surv. Geophys.*, vol. 21, no. 2, pp. 147–157, 2000.

[13] S. Xiao et al., "MoCG: Modality characteristics-guided semantic segmentation in multimodal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5625818.

[14] I. Misra, M. K. Rohil, S. Manthira Moorthi, and D. Dhar, "Feature based remote sensing image registration techniques: A comprehensive and comparative review," *Int. J. Remote Sens.*, vol. 43, no. 12, pp. 4477–4516, Jun. 2022.

[15] F. Xu, Y. Shi, P. Ebel, W. Yang, and X. X. Zhu, "Multimodal and multiresolution data fusion for high-resolution cloud removal: A novel baseline and benchmark," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5600615.

[16] F. Xu, Y. Shi, W. Yang, G.-S. Xia, and X. X. Zhu, "CloudSeg: A multimodal learning framework for robust land cover mapping under cloudy conditions," *ISPRS J. Photogramm. Remote Sens.*, vol. 214, pp. 21–32, Aug. 2024.

[17] V. Sarukkai, A. Jain, B. Uzkent, and S. Ermon, "Cloud removal in satellite images using spatiotemporal generative networks," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2020, pp. 1785–1794.

[18] J. Gawlikowski, P. Ebel, M. Schmitt, and X. X. Zhu, "Explaining the effects of clouds on remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9976–9986, 2022.

[19] R. S. Stanković and B. J. Falkowski, "The Haar wavelet transform: Its status and achievements," *Comput. Electr. Eng.*, vol. 29, no. 1, pp. 25–44, Jan. 2003.

[20] H. Wang, Z. Zhang, Z. Hu, and Q. Dong, "SAR-to-optical image translation with hierarchical latent features," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5233812.

[21] Y. Wang and M. Li, "Urban impervious surface detection from remote sensing images: A review of the methods and challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 3, pp. 64–93, Sep. 2019.

[22] F. Li, D. A. Clausi, L. Xu, and A. Wong, "ST-IRGS: A region-based self-training algorithm applied to hyperspectral image classification and segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 3–16, Jan. 2018.

[23] T. Kasetkasem, P. Rakwatin, R. Sirisommai, and A. Eiumnoh, "A joint land cover mapping and image registration algorithm based on a Markov random field model," *Remote Sens.*, vol. 5, no. 10, pp. 5089–5121, Oct. 2013.

[24] Z. Xu, W. Jiang, and J. Geng, "Texture-aware causal feature extraction network for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5103512.

[25] X. Ma, X. Zhang, M.-O. Pun, and M. Liu, "A multilevel multimodal fusion transformer for remote sensing semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5403215.

[26] W. Xie, Y. Lu, D. Li, and Y. Li, "Ebbinghaus-curve guided low-rank component-induced attention for multisource remote sensing classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5644712.

[27] A. T. Fuller, K. Millard, and J. R. Green, "CROMA: Remote sensing representations with contrastive radar-optical masked autoencoders," in *Proc. 37th Int. Conf. Neural Inf. Process. Syst.*, 2023, pp. 5506–5538.

[28] Y. Wang, B. Zhang, W. Zhang, D. Hong, B. Zhao, and Z. Li, "Cloud removal with SAR-optical data fusion using a unified spatial–spectral residual network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5600820.

[29] Y. Li, F. Wei, Y. Zhang, W. Chen, and J. Ma, "HS2P: Hierarchical spectral and structure-preserving fusion network for multimodal remote sensing image cloud and shadow removal," *Inf. Fusion*, vol. 94, pp. 215–228, Jun. 2023.

[30] F. Xu et al., "GLF-CR: SAR-enhanced cloud removal with global–local fusion," *ISPRS J. Photogramm. Remote Sens.*, vol. 192, pp. 268–278, Oct. 2022.

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[31] J. Ma et al., "SCT-CR: A synergistic convolution-transformer modeling method using SAR-optical data fusion for cloud removal," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 130, Jun. 2024, Art. no. 103909.
[32] W. Zhang, J. Mei, and Y. Wang, "DMDiff: A dual-branch multimodal conditional guided diffusion model for cloud removal through SAR-optical data fusion," *Remote Sens.*, vol. 17, no. 6, p. 965, Mar. 2025.
[33] S. Ruder, "An overview of multi-task learning in deep neural networks," 2017, *arXiv:1706.05098*.
[34] Y. Cai, Y. Yang, Y. Shang, Z. Shen, and J. Yin, "DASRSNet: Multitask domain adaptation for super-resolution-aided semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5600218.
[35] L. Zhao, Y. Hu, X. Yang, Z. Dou, and L. Kang, "Robust multi-task learning network for complex LiDAR point cloud data preprocessing," *Expert Syst. Appl.*, vol. 237, Mar. 2024, Art. no. 121552.
[36] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Álvarez, and P. Luo, "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. 35th Int. Conf. Neural Inf. Process. Syst.*, 2021, pp. 12077–12090.
[37] Y. Yang, G. Yuan, and J. Li, "SFFNet: A wavelet-based spatial and frequency domain fusion network for remote sensing segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 3000617.
[38] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19.
[39] Q. Hou, D. Zhou, and J. Feng, "Coordinate attention for efficient mobile network design," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13708–13717.
[40] J. Pan et al., "HDRSA-Net: Hybrid dynamic residual self-attention network for SAR-assisted optical image cloud and shadow removal," *ISPRS J. Photogramm. Remote Sens.*, vol. 218, pp. 258–275, Dec. 2024.
[41] X. Li et al., "MCANet: A joint semantic segmentation framework of optical and SAR images for land use classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 106, Feb. 2022, Art. no. 102638.
[42] G. Li, Y. Wang, Z. Liu, X. Zhang, and D. Zeng, "RGB-T semantic segmentation with location, activation, and sharpening," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 3, pp. 1223–1235, Mar. 2023.
[43] X. Fan, W. Zhou, X. Qian, and W. Yan, "Progressive adjacent-layer coordination symmetric cascade network for semantic segmentation of multimodal remote sensing images," *Expert Syst. Appl.*, vol. 238, Mar. 2024, Art. no. 121999.
[44] R. Liu, J. Ling, and H. Zhang, "SoftFormer: SAR-optical fusion transformer for urban land use and land cover classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 218, pp. 277–293, Dec. 2024.
[45] J. Zhang, H. Liu, K. Yang, X. Hu, R. Liu, and R. Stiefelhagen, "CMX: Cross-modal fusion for RGB-X semantic segmentation with transformers," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 12, pp. 14679–14694, Dec. 2023.



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